

1.

An algorithm runs in $5n^2 + 3n + 10$ steps.

☞ Find Big-O complexity

2.

If input size doubles, an $O(n^2)$ algorithm increases time by:

3.

An algorithm takes $\log_2(n)$ steps. For $n = 1024$, steps = ?

4.

Recursive relation: $T(n) = T(n-1) + 1$

☞ Solve complexity

5.

Recursive relation: $T(n) = 2T(n/2) + n$

☞ Complexity

1.

Given:

$$T(n) = 5n^2 + 3n + 10$$

☞ In Big-O, we ignore constants and lower-order terms.
So dominant term = n^2

Answer: $O(n^2)$

2.

Given: $O(n^2)$

If input doubles:

$$T(n) = n^2$$

$$T(2n) = (2n)^2 = 4n^2$$

☞ Time increases by **4 times**

Answer: 4 times

3.

Given: $\log_2(n)$, $n = 1024$

We know:

$$1024 = 2^{10}$$

$$\text{So, } \log_2(1024) = 10$$

Answer: 10 steps

4.

Given recurrence:

$$T(n) = T(n-1) + 1$$

Expanding:

$$T(n) = T(n-2) + 2$$

$$T(n) = T(n-3) + 3$$

...

$$T(n) = T(1) + (n-1)$$

☞ Growth is linear

Answer: $O(n)$

5.

Given recurrence:

$$T(n) = 2T(n/2) + n$$

Using Master Theorem:

$$a = 2, b = 2, f(n) = n$$

$$n^{(\log_b a)} = n^{(\log_2 2)} = n^1 = n$$

So, Case 2 applies:

$$T(n) = O(n \log n)$$

Answer: $O(n \log n)$